

Matrices

4.4 Elementary Row Transformation

Uses:

- (a) Inverse
- (b) Linear Equation
- (c) Rank

Allowed Operation:

- (a) Interchange of rows
- (b) Addition of a constant multiple of one row to other
- (c) Multiplication of an equation by a non-zero constant.

$$\begin{aligned}\Rightarrow 3x_1 + 2x_2 + 2x_3 - 5x_4 &= 8 \\ 0.6x_1 + 1.5x_2 + 1.5x_3 - 5.4x_4 &= 2.7 \\ 1.2x_1 - 0.2x_2 - 0.3x_3 + 2.4x_4 &= 2.1\end{aligned}$$

$$\Rightarrow \text{Let } A = \begin{bmatrix} 3 & 2 & 2 & -5 & 8 \\ 0.6 & 1.5 & 1.5 & -5.4 & 2.7 \\ 1.2 & -0.2 & -0.3 & 2.4 & 2.1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 3 & 2 & 2 & -5 & 8 \\ 0.6 & 1.1 & 1.1 & -4.4 & 1.1 \\ 0 & -1 & -1.1 & 4.4 & -1.1 \end{bmatrix}, \begin{pmatrix} R_2 \rightarrow R_2 - 0.2R_1 \\ R_3 \rightarrow R_3 - 0.4R_1 \end{pmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 3 & 2 & 2 & -5 & 8 \\ 0.6 & 1.1 & 1.1 & -4.4 & 1.1 \\ 0.6 & 0.1 & 0 & 0 & 0 \end{bmatrix}, (R_3 \rightarrow R_3 + R_2)$$